A.4 Details on Super-pixel Experiments

We provide an example of the rotated images and corresponding super-pixel graphs in Fig. 6. (Note that classes "6" and "9" may be confused under severe distribution shift, i.e. 90 degrees rotation or more. Hence, to avoid harming class information, our experiments only consider distribution shift from rotation up to 40 degrees.)

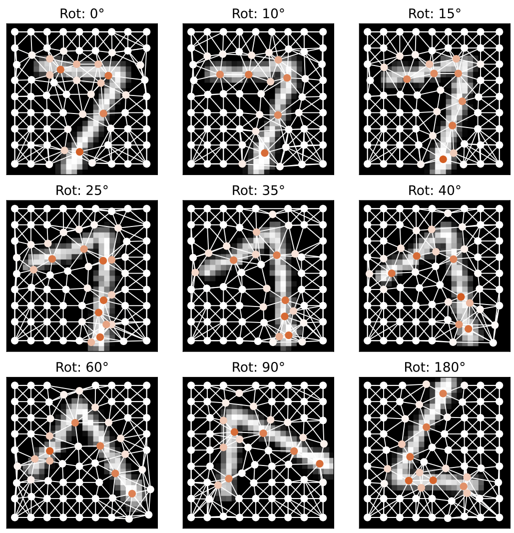

Figure 6: **Rotated Super-pixel MNIST.** Rotating images prior to creating super-pixels to leads to some structural distortion (Ding et al., 2021). However, we can see that the class-discriminative information is preserved, despite rotation. This allows for simulating different levels of graph structure distribution shifts, while still ensuring that samples are valid.

Tables 4 and 5 provided expanded results on the rotated image super-pixel graph classification task, discussed in Sec. 6.1.

In Table 7 we focus on the calibration results on this task for GPS variants alone. Across all levels of distribution shift, the best method is our strategy for applying G-ΔUQ to a pretrained model– demonstrating that this is not just a practical choice when it is infeasible to retrain a model, but can lead to powerful performance by any measure. Second-best on all datasets is applying G-ΔUQ during training, further highlighting the benefits of stochastic anchoring.

In addition to the structural distribution shifts we get by rotating the images before constructing super-pixel graphs, we also simulate feature distribution shifts by adding Gaussian noise with different standard deviations to the pixel value node features in the super-pixel graphs. In Table 8, we report accuracy and calibration results for varying levels of distribution shift (represented by the size of the standard deviation of the Gaussian noise). Across different levels of feature distribution shift, we also see that G-ΔUQ results in superior calibration, while maintaining competitive or in many cases superior accuracy.

Table 4: **RotMNIST-Accuracy.** Here, we report expanded results (accuracy) on the Rotated MNIST dataset, including a variant that combines G-ΔUQ with Deep Ens. Notably, we see that anchored ensembles outperform basic ensembles in both accuracy and calibration. (Best results for models using Deep Ens. and those not using it marked separately.)

MODEL	G-ΔUQ?	LPE?	Avg. Test (↑)	Acc. (10) (↑)	Acc. (15) (↑)	Acc. (25) (↑)	Acc. (35) (↑)	Acc. (40) (↑)
GatedGCN	✗	✗	0.947 ±0.002	0.918 ±0.002	0.904 ±0.005	0.828 ±0.009	0.738 ±0.009	0.679 ±0.007
	✓	✗	0.933 ±0.015	0.894 ±0.019	0.878 ±0.020	0.794 ±0.032	0.698 ±0.036	0.636 ±0.048
	✗	✓	0.949 ±0.002	0.917 ±0.004	0.904 ±0.005	0.829 ±0.007	0.744 ±0.007	0.685 ±0.006
	✓	✓	0.915 ±0.032	0.872 ±0.038	0.852 ±0.0414	0.776 ±0.039	0.680 ±0.037	0.631 ±0.033
GPS	✗	✓	**0.970** ±0.001	**0.948** ±0.001	**0.938** ±0.001	**0.873** ±0.006	**0.770** ±0.013	**0.688** ±0.009
	✓	✓	0.969 ±0.001	0.946 ±0.003	0.937 ±0.003	0.869 ±0.003	0.769 ±0.012	0.679 ±0.014
GPS (Pretrained)	✓	✓	0.967 ±0.002	0.945 ±0.004	0.934 ±0.005	0.864 ±0.009	0.759 ±0.010	0.674 ±0.002
GatedGCN-DENS	✗	✗	0.963 ±0.0002	0.943 ±0.001	0.933 ±0.001	0.874 ±0.002	0.794 ±0.002	0.731 ±0.002
	✓	✗	0.949 ±0.008	0.922 ±0.008	0.907 ±0.011	0.828 ±0.020	0.733 ±0.032	0.662 ±0.046
	✗	✓	0.965 ±0.001	0.943 ±0.001	0.933 ±0.001	0.873 ±0.001	0.792 ±0.004	0.736 ±0.003
	✓	✓	0.954 ±0.005	0.930 ±0.010	0.917 ±0.011	0.850 ±0.023	0.759 ±0.025	0.696 ±0.032
GPS-DENS	✗	✓	**0.980** ±0.000	**0.969** ±0.000	**0.961** ±0.000	**0.913** ±0.000	**0.834** ±0.000	**0.750** ±0.000
	✓	✓	0.978 ±0.001	0.963 ±0.000	0.953 ±0.001	0.905 ±0.000	0.822 ±0.002	0.736 ±0.003

Table 5: **RotMNIST-Calibration.** Here, we report expanded results (calibration) on the Rotated MNIST dataset, including a variant that combines G-ΔUQ with Deep Ens. Notably, we see that anchored ensembles outperform basic ensembles in both accuracy and calibration. (Best results for models using Deep Ens. and those not using it marked separately.)

MODEL	G-ΔUQ	LPE?	Avg.ECE (↓)	ECE (10) (↓)	ECE (15) (↓)	ECE (25) (↓)	ECE (35) (↓)	ECE (40) (↓)
GatedGCN-TS	✗	✗	0.035 ±0.001	0.054 ±0.002	0.062 ±0.003	0.118 ±0.007	0.185 ±0.006	0.233 ±0.008
	✗	✓	0.033 ±0.002	0.053 ±0.002	0.061 ±0.004	0.116 ±0.005	0.179 ±0.006	0.225 ±0.005
GatedGCN	✗	✗	0.038 ±0.001	0.059 ±0.001	0.068 ±0.340	0.126 ±0.008	0.195 ±0.012	0.245 ±0.011
	✓	✗	**0.018** ±0.008	0.029 ±0.013	**0.033** ±0.164	0.069 ±0.033	0.117 ±0.048	0.162 ±0.067
	✗	✓	0.036 ±0.003	0.059 ±0.002	0.068 ±0.340	0.125 ±0.006	0.191 ±0.007	0.240 ±0.008
	✓	✓	0.022 ±0.007	**0.028** ±0.014	0.034 ±0.169	**0.062** ±0.022	**0.109** ±0.019	**0.141** ±0.019
GPS-TS	✗	✓	0.024 ±0.001	0.041 ±0.001	0.049 ±0.001	0.102 ±0.006	0.188 ±0.012	0.261 ±0.008
GPS	✗	✓	0.026 ±0.001	0.044 ±0.001	0.052 ±0.156	0.108 ±0.006	0.197 ±0.012	0.273 ±0.008
	✓	✓	0.022 ±0.001	0.037 ±0.005	0.044 ±0.133	0.091 ±0.008	0.165 ±0.018	0.239 ±0.018
GPS (Pretrained)	✓	✓	0.021 ±0.001	0.032 ±0.003	0.039 ±0.116	0.083 ±0.002	0.153 ±0.007	0.217 ±0.012
GatedGCN-DENS	✗	✗	0.026 ±0.000	0.038 ±0.001	0.042 ±0.001	0.084 ±0.002	0.135 ±0.001	0.185 ±0.003
	✓	✗	**0.014** ±0.003	**0.018** ±0.005	**0.021** ±0.005	0.036 ±0.012	0.069 ±0.032	0.114 ±0.056
	✗	✓	0.024 ±0.001	0.038 ±0.001	0.043 ±0.002	0.083 ±0.001	0.139 ±0.004	0.181 ±0.002
	✓	✓	0.017 ±0.002	0.024 ±0.005	0.027 ±0.008	**0.030** ±0.004	**0.036** ±0.012	**0.059** ±0.033
GPS-DENS	✗	✓	0.016 ±0.001	0.026 ±0.002	0.030 ±0.000	0.066 ±0.000	0.123 ±0.000	0.195 ±0.000
	✓	✓	**0.014** ±0.000	0.023 ±0.002	0.027 ±0.003	0.055 ±0.004	0.103 ±0.006	0.164 ±0.006

Table 6: **Accuracy of GPS Variants on RotatedMNIST.** We focus on the accuracy results for GPS variants on rotated MNIST dataset. Using G-ΔUQ (with or without pretraining) remains close in accuracy to foregoing it, generally within the range of the standard deviation of the results.

MODEL	G-ΔUQ?	Avg. Test (↑)	Acc. (10) (↑)	Acc. (15) (↑)	Acc. (25) (↑)	Acc. (35) (↑)	Acc. (40) (↑)
GPS	✗	**0.970** ±0.001	**0.948** ±0.001	**0.938** ±0.001	**0.873** ±0.006	**0.770** ±0.013	**0.688** ±0.009
	✓	0.969 ±0.001	0.946 ±0.003	0.937 ±0.003	0.869 ±0.003	0.769 ±0.012	0.679 ±0.014
GPS (Pretrained)	✓	0.967 ±0.002	0.945 ±0.004	0.934 ±0.005	0.864 ±0.009	0.759 ±0.010	0.674 ±0.002

Table 7: **Calibration of GPS Variants on RotatedMNIST.** We focus on the calibration results for GPS variants on rotated MNIST dataset. Across the board, we see improvements from using G-ΔUQ , with our strategy of applying it to a pretrained model doing best.

MODEL	G-ΔUQ	Avg.ECE (↓)	ECE (10) (↓)	ECE (15) (↓)	ECE (25) (↓)	ECE (35) (↓)	ECE (40) (↓)
GPS-TS	✗	0.024 ±0.001	0.041 ±0.001	0.049 ±0.001	0.102 ±0.006	0.188 ±0.012	0.261 ±0.008
GPS	✗	0.026 ±0.001	0.044 ±0.001	0.052 ±0.156	0.108 ±0.006	0.197 ±0.012	0.273 ±0.008
	✓	0.022 ±0.001	0.037 ±0.005	0.044 ±0.133	0.091 ±0.008	0.165 ±0.018	0.239 ±0.018
GPS (Pretrained)	✓	**0.021** ±0.001	**0.032** ±0.003	**0.039** ±0.116	**0.083** ±0.002	**0.153** ±0.007	**0.217** ±0.012

Table 8: **MNIST Feature Shifts**. G-ΔUQ improves calibration and maintains competitive or even improved accuracy across varying levels of feature distribution shift.

				STD = 0.1		STD = 0.2		STD = 0.3		STD = 0.4	
MODEL	LPE?	G-ΔUQ?	Calibration	Accuracy (↑)	ECE (↓)	Accuracy (↑)	ECE (↓)	Accuracy (↑)	ECE (↓)	Accuracy (↑)	ECE (↓)
GatedGCN	×	×	×	0.742±0.005	0.186±0.018	0.481±0.015	0.414±0.092	0.293±0.074	0.606±0.147	0.197±0.092	0.71±0.178
	×	✓	×	**0.773**±0.053	**0.075**±0.032	0.536±0.010	**0.160**±0.087	**0.356**±0.101	0.422±0.083	**0.249**±0.074	**0.529**±0.047
	✓	×	×	0.751±0.02	0.176±0.014	0.519±0.004	0.348±0.03	0.345±0.032	0.485±0.096	0.233±0.043	0.581±0.142
	✓	✓	×	0.745±0.026	0.100±0.036	**0.541**±0.040	0.235±0.067	**0.355**±0.062	**0.408**±0.116	0.242±0.063	0.539±0.139